The Chinese Postman Problem — 2026 Revision

Revised edition of the 1997 final-year project by Miquel Monserrat Pons (supervisor: Ramón Companys Pascual)

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The 2026 revision was prepared with AI assistance (Claude, by Anthropic). The 1997 project was not.

Abstract

This report revises a 1997 final-year project on the Chinese Postman Problem: find the cheapest closed route that covers every street of a network at least once.

The method is unchanged in outline: pair the odd-degree junctions by shortest paths at minimum total cost, duplicate those paths, and walk the resulting Eulerian circuit. The 1997 branch-and-bound for the pairing step is replaced by Edmonds' blossom algorithm, which is exact and polynomial; shortest paths use Dijkstra, the circuit uses Hierholzer's linear-time algorithm, the directed version is added through minimum-cost flow, and the mixed version, with one-way and two-way streets together and NP-hard, through branch-and-cut and a heuristic built for this project. The report then follows the paths left open in 1997: streets whose first pass costs more than the next, streets that need no service (the rural postman problem) and several postmen sharing a district, each with its complexity, its method and the measured distance from the optimum. It adds the windy postman, where each direction of a street has its own cost, and carries the rural and several-postmen problems to directed, mixed and windy networks.

The revision combines certified sparse matching for large networks, dual-guided local search for mixed routing, and a unified open-source implementation with a browser-based solver. These were checked by measurement, not against the whole literature, so they are not claims of priority.

The Python implementation was tested on classic graphs, on 92 benchmark networks from the arc-routing literature (including real road networks in Lancashire and Flanders), and on synthetic towns of up to 40,000 junctions. For the undirected CPP, every tested instance was solved to proven optimality. With a second backend that runs the LEMON library on a sparse candidate graph, adds pairs by dual pricing and checks a dual certificate, a network with 12,066 odd vertices solves in a median 110 seconds, against about 750 odd vertices in the same time in pure Python, and one with 19,942 in 250 seconds. On 63 benchmarks where every street needs service, the single-postman tour costs a median 95.1% of the best known multi-vehicle solution, but as little as 47.5% when capacity is scarce: the CPP is a close bound for fleet routing only while vehicles have capacity to spare.

For mixed networks, branch-and-cut proves the optimum on every benchmark network within 5.8 seconds, on all 20 towns of up to 400 junctions and on 8 of the 10 towns of 900 and 1,600 junctions, and on 17 of a sample of 30 published instances of 500 to 3,000 junctions within two minutes. All 39 optima proved on published instances equal the published values, and five real street networks from OpenStreetMap, of up to 1,154 junctions, are proved in at most 34 seconds. The heuristic averages 0.12% above the optimum on the benchmarks, 1.35% on the towns and 2.09% on that sample, with a worst case of 5.8% where few streets are one-way.

On published instances, Win's method for the windy postman averages 0.10% above 99 optima, with a worst case of 0.44%. The best rural route averages 0.62% above 47 windy optima, with a worst case of 7.6%, and branch-and-cut proves 6 of 12 mixed rural instances in five minutes and 113 of 150 related directed ones, of up to 60,000 arcs, each equal to the published value. The several-postmen heuristic averages 2.15% above 629 published optima, with a worst case of 17.2%, assuming a depot that the files do not state. On two-way networks, splitting an optimal route carries the factor 2 − 1/k of section 10.4. For the generalized several-postmen problem on asymmetric networks, no general approximation ratio has been proved here.

For the CPP itself, a fast answer and a proven one are now the same answer. For its NP-hard extensions, an exact search goes as far as it can, and beyond that a heuristic comes with a lower bound that certifies how far its route can be from the optimum. A JavaScript port of the solver runs in the web browser, so anyone can solve their own street network, from a file or straight from OpenStreetMap, at miquelmp.github.io/chinese-postman-problem.

1. Introduction

The Chinese Postman Problem (CPP) asks for the cheapest closed walk that traverses every edge of a graph at least once. The 1997 project solved it by duplicating edges until every vertex has even degree, choosing those duplicates with a custom branch-and-bound (B&B) search. This revision keeps the same overall method but replaces the B&B with an exact, polynomial-time matching algorithm, and tests the result on a much wider set of instances. The polynomial algorithms used here were all published before 1997; that project set out to study a branch-and-bound alternative. What has changed since is the engineering: implementations such as Blossom V and LEMON, compiled libraries and integer-programming solvers reach far larger networks.

The problem is named after the Chinese mathematician Kwan Mei-Ko (Guan Meigu), who posed it in 1960. Edmonds and Johnson showed in 1973 that the undirected version can be solved in polynomial time by reducing it to a minimum-weight perfect matching.

Typical applications are any service that must cover streets rather than visit points:

- mail and parcel delivery on foot;
- snow ploughing and gritting;
- refuse collection and street sweeping;
- park and street-tree maintenance crews;
- inspection of pipelines, power lines and road markings.

What changed since 1997

Aspect	1997 project	2026 revision
Matching step	Binary B&B over edges, lower bound from row minima	Edmonds' blossom algorithm (exact, polynomial)
Shortest paths	Bellman-type iteration on the full V × V matrix (the 1997 report's "Fulkerson algorithm")	Dijkstra from each odd vertex only
Eulerian circuit	Hierholzer-style cycle splicing with linear scans of adjacency queues	Hierholzer's algorithm with one pointer per vertex, O(m)
Directed graphs	Proposed as future work	Added, by minimum-cost flow (Edmonds and Johnson, 1973)
Language and platform	Turbo C, MS-DOS, 640 KB memory model	Python 3 with NetworkX, any platform

Aspect	1997 project	2026 revision
Practical limit	About 20 odd vertices to proven optimality	About 12,000 odd vertices (40,000 junctions) in under two minutes with LEMON; about 750 in pure Python
Mixed streets (one-way and two-way)	Not covered	Exact branch-and-cut (HiGHS) and a heuristic with local search

Contributions of this revision

1. A clean, tested reimplementation of the full CPP pipeline.
2. A computational study on nine families of instances, none taken from the original Vilanova i la Geltrú case studies.
3. A comparison of the 1997 branch-and-bound with the exact method, and a tightened lower bound for anyone who extends the B&B to harder variants.
4. The paths left open in 1997, implemented and measured: directed and mixed networks, service and travel costs, the rural postman problem, several postmen and capacitated fleets.

What is established and what is new

Most of the methods here are established, and this report does not claim them. Edmonds' algorithm, minimum-cost flow, Hierholzer's algorithm, Frederickson's heuristics, Ulusoy's split, price-and-repair for sparse matching (Cook and Rohe, 1999) and LEMON come from the works cited. The models of chapters 9 and 10 are standard or follow the sources named there.

Three things were built for this project: the certified route to large matchings, with a dual certificate checked on every run (section 4.4), the dual-guided local search for mixed networks (section 9.3), and the combination of known parts into one open package with a browser version. They were checked by measurement, not against the whole literature, so they are not claims of priority.

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2. Graph-theoretic background

Only four ideas are needed: graphs with edge costs, vertex degree, Euler's theorem and matchings. The 1997 report spent a chapter on general nomenclature; this revision keeps just what the method uses.

Graph. G = (V, E) has a vertex set V and an edge set E, with n = |V| and m = |E|. In an undirected graph each edge joins two vertices without orientation; in a directed graph each arc goes from a tail to a head. A multigraph allows several parallel edges between the same pair. Each edge e carries a cost c(e) > 0, such as length, time or fuel.

Degree. The degree of a vertex is the number of edge ends that meet it. Because every edge has two ends, the sum of all degrees is 2m, so every graph has an even number of odd-degree vertices. This handshake lemma is why the odd vertices can always be paired up.

Walks and circuits. A walk is a sequence of vertices and edges, starting and ending at a vertex, in which each edge joins the vertex before it to the vertex after it. A closed walk returns to its start. An Eulerian circuit is a closed walk that uses every edge exactly once.

Euler's theorem. A connected multigraph has an Eulerian circuit if and only if every vertex has even degree. For a strongly connected directed multigraph, the condition is that every vertex has in-degree equal to out-degree. Euler (1741) proved that even degrees are necessary and stated the converse without proof; Hierholzer (1873) published the first proof of it.

Matchings. A matching is a set of edges no two of which share a vertex. It is perfect when it covers every vertex. Given edge costs, a minimum-weight perfect matching is a perfect matching of least total cost. On a complete graph with an even number of vertices a perfect matching always exists.

Shortest paths. d(u, v) is the cost of a cheapest path from u to v. With non-negative costs it is computed by Dijkstra's algorithm (1959) per source in O((n + m) log n) time with a binary heap, or O(m + n log n) with a Fibonacci heap.

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3. The Chinese Postman Problem

The optimal postman tour costs the sum of all edges plus the cheapest way to pair up the odd vertices by shortest paths. Everything else in this report is about computing that pairing and then walking the resulting graph.

3.1 Formulation

Given a connected undirected multigraph G = (V, E) with costs c(e) > 0, find a closed walk that traverses every edge at least once and has minimum total cost. Equivalently, choose how many extra copies x(e) ≥ 0 of each edge to add so that the augmented multigraph is Eulerian, at minimum added cost:

$$\min \sum_{e \in E} c(e)\, x(e) \quad \text{s.t.} \quad \deg_G(v) + \sum_{e \ni v} x(e) \equiv 0 \pmod 2 \ \ \forall v \in V, \qquad x(e) \in \mathbb{Z}_{\geq 0}$$

An optimal solution never uses more than one extra copy of an edge, so x(e) ∈ {0, 1}. The set of duplicated edges is a minimum-cost T-join, where T is the set of odd vertices: a set of edges in which exactly the vertices of T have odd degree.

3.2 Edmonds' method

1. Let T be the set of odd-degree vertices. If T is empty, G is already Eulerian; go to step 5.
2. Compute shortest-path distances d(u, v) for every pair u, v in T.
3. Build the complete graph K_T on T with edge costs d(u, v), and find a minimum-weight perfect matching M* on it.
4. For each pair (u, v) in M*, duplicate every edge on a shortest u–v path in G.
5. The augmented multigraph has all degrees even; find an Eulerian circuit in it. That circuit is an optimal postman tour.

The total cost is c(E) + c(M*). The method is exact for the following reason. Any set of duplicated edges that makes all degrees even is a T-join, and every T-join contains edge-disjoint paths that pair up the vertices of T. Each of those paths costs at least the shortest-path distance between its ends, so no T-join costs less than c(M*). The paths chosen in step 4 reach that value: with positive costs they cannot share an edge, because removing both copies of a shared edge would keep every parity and give a T-join cheaper than c(M*).

3.3 Worked example from 1997

The 1997 report used a small park-maintenance network with 8 vertices and 12 edges. Its edge costs, taken from the street plan, are:

Edge	1–8	1–7	8–7	1–4	7–4	7–6	4–5	6–5	6–3	3–5	5–2	3–2
Cost	2	1	5	6	4	7	2	1	4	7	6	9

As in the original report, costs stand for travel time or fuel rather than plain distance.

The odd vertices are T = {1, 3, 4, 6}. The three possible perfect matchings on K_T cost:

Matching	Cost
(1,3) + (4,6)	12 + 3 = 15
(1,4) + (3,6)	5 + 4 = 9
(1,6) + (3,4)	8 + 7 = 15

The optimum is M* = {(1,4), (3,6)}. It duplicates the path 1–7–4 and the edge 3–6. The original network costs 54, so the optimal tour costs 54 + 9 = 63 and has 15 edges. The 2026 code reproduces this exactly and returns the tour 1–8–7–1–7–4–7–6–3–2–5–3–6–5–4–1.

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4. Minimum-weight perfect matching

The matching step is solved exactly, in polynomial time, by Edmonds' blossom algorithm. This replaces the 1997 branch-and-bound entirely: there is no longer any instance size at which the method has to fall back on a time limit or a heuristic.

4.1 Linear programming formulation

Let K = (T, F) be the complete graph on the odd vertices, with cost d(e) on each edge, and let x(e) = 1 when edge e is in the matching. For any vertex set S, δ(S) is the set of edges with exactly one end in S. Edmonds (1965) showed that the following linear program has an integral optimum, so it describes perfect matchings exactly:

$$\min \sum_{e \in F} d(e)\, x(e) \quad \text{s.t.} \quad x(\delta(v)) = 1 \;\; \forall v \in T, \qquad x(\delta(S)) \geq 1 \;\; \forall S \subseteq T,\ |S| \geq 3 \text{ odd}, \qquad x \geq 0$$

Without the odd-set constraints the program can return half-integral solutions on odd cycles: three vertices joined in a triangle, each edge at 0.5. The odd-set constraints rule these out. There are exponentially many of them, but the algorithm never lists them; it only touches the few that matter.

4.2 How the blossom algorithm works

The algorithm is a primal-dual method. It keeps a matching, which starts empty, and a feasible dual solution: a price y(v) on each vertex and a price z(B) on each odd set B in use. An edge is tight when the prices of its ends, plus those of the odd sets it crosses, equal its cost. The algorithm repeats three moves until every vertex is matched:

1. **Grow.** From an unmatched vertex, build an alternating tree using tight edges only. If it reaches another unmatched vertex, flip the path between them: the matching grows by one edge.
2. **Shrink.** If the tree closes an odd cycle (a blossom), contract the cycle into a single pseudo-vertex and keep growing.
3. **Adjust prices.** If the tree is stuck, change the dual prices by the largest step that keeps them feasible. This makes at least one new edge tight, or lets a pseudo-vertex be expanded again.

When every vertex is matched, the matching and the prices satisfy complementary slackness, which proves optimality. Chapter 3 of the 1997 report already presented this algorithm. The 2026 version uses it directly, through a mature library implementation, so the matching no longer needs a purpose-built search.

4.3 Complexity and implementations

Implementation	Language	Worst-case time	Notes
Edmonds (1965)	—	$O(n^4)$	Original algorithm
Gabow (1973), Lawler (1976)	—	$O(n^3)$	Standard textbook bound
Galil, Micali and Gabow (1986)	—	$O(nm \log n)$	Better on sparse graphs
Blossom V, Kolmogorov (2009)	C++	Believed $O(n^3m)$, no better than Blossom IV; much faster in practice	Used in earlier versions; non-commercial licence
LEMON, Dezső, Jüttner and Kovács (2011)	C++	$O(nm \log n)$	Fast backend in this revision; Boost licence
NetworkX `min_weight_matching`	Python	$O(n^3)$	Default backend in this revision

Here n is the number of odd vertices and m the number of edges of K, so $m = n(n - 1)/2$. NetworkX was chosen for readability and portability: the undirected and directed solvers take 133 lines of Python, and the Eulerian circuit 42 more. Its pure-Python matching becomes the

bottleneck above a few hundred odd vertices (section 7.4), so large networks use a second backend built on LEMON, described next.

4.4 Large networks: a sparse candidate graph with dual pricing

The complete graph K has about $n^2/2$ edges, yet an optimal matching almost always pairs each odd vertex with one of its nearest odd neighbours. The fast backend exploits this without giving up the optimality guarantee:

1. Compute shortest-path distances from every odd vertex with SciPy's compiled Dijkstra.
2. Build a sparse candidate graph: each odd vertex's 10 nearest odd vertices, plus the pairs of a greedy matching so that a perfect matching is sure to exist.
3. Solve it with LEMON's C++ blossom code and read the optimal dual prices y.
4. Pricing: scan all pairs (u, v) not in the graph and add those with negative reduced cost, d(u, v) < y(u) + y(v), counting blossom prices. Re-solve and repeat.
5. When no pair has negative reduced cost, the duals are feasible for the whole complete graph, so the matching is optimal for it too.

This is column generation applied to matching, the strategy of Cook and Rohe (1999), which Blossom V also uses for geometric instances. On the test networks the first candidate graph comes very close: at most 83 pairs had to be added, in up to two pricing rounds, followed by a last round that confirms optimality. Related ideas reach even further: the sparse blossom algorithm of Higgott and Gidney (2025), developed for quantum error correction, matches millions of vertices without ever building distances between all pairs.

Earlier versions used Blossom V at this step. It may be used for evaluation and non-commercial research only and may not be redistributed, so a reader could not freely repeat those runs. LEMON has a permissive licence (Boost), so the part it needs ships with the package and is compiled by one script. NetworkX stays the default when it has not been built, or when costs are not integers.

A result should be checkable without trusting the solver. Every run of the fast backend therefore ends by checking a certificate: vertex prices and blossom prices such that, with w = C − d,

$$u_i + u_j + \sum_{B \ni i,j} z_B \geq w_{ij} \;\; \forall i,j, \qquad z_B \geq 0, \qquad w(M) = \sum_i u_i + \sum_B z_B \, \lfloor |B|/2 \rfloor \, .$$

By weak duality no perfect matching on the complete graph has a higher weight, so M is optimal. The check needs only the distances and these numbers, not the solver that produced them. As a second test, `tools/certify_matching.py` repeats the whole route with the JavaScript blossom code of section 6.6.

4.5 Milestones in matching, 1965–2026

Minimum-weight perfect matching has kept improving since Edmonds' blossom algorithm, both in theory and in working code. The table lists the milestones most relevant to postman problems, dated by publication.

Year	Authors	Contribution	Relevance here
1965	Edmonds	Blossom algorithm and the matching polytope	Foundation of the exact method, $O(n^4)$
1973–1976	Gabow; Lawler	$O(n^3)$ implementations	Textbook bound, also that of NetworkX
1980	Micali and Vazirani	$O(m\sqrt{n})$ maximum-cardinality matching	Fastest bound for the unweighted case
1986	Galil, Micali and Gabow	$O(nm \log n)$ weighted matching	Faster on sparse graphs
1990	Gabow	$O(nm + n^2 \log n)$	Still the best strongly polynomial bound
1991	Gabow and Tarjan	Cost scaling for integer weights	Near $O(m\sqrt{n} \log(nN))$
1999	Cook and Rohe	Blossom IV, with variable ε and pricing	Solved geometric instances of up to 5 million points
2002	Mehlhorn and Schäfer	$O(nm \log n)$ implementation in LEDA	Data structures that make the bound practical
2009	Kolmogorov	Blossom V	Used in earlier versions of this package
2011	Dezső, Jüttner and Kovács	LEMON graph library	Fast backend of this revision
2014	Duan and Pettie	$(1 - \varepsilon)$-approximate weighted matching in linear time	For maximum-weight matching; gives no guarantee for a minimum-cost perfect pairing
2018	Duan, Pettie and Su	Exact scaling algorithm, $O(m\sqrt{n} \log(nN))$	Best known bound for integer costs
2022	Chen, Kyng, Liu, Peng, Probst Gutenberg and Sachdeva	Minimum-cost flow in almost-linear time	Directed CPP in almost-linear time, in theory

Year	Authors	Contribution	Relevance here
2023	Wu and Zhong	Fusion Blossom, parallel matching	Quantum error correction at scale
2025	Higgott and Gidney	Sparse Blossom	Millions of vertices without all-pair distances
2026	Arkhipov and Kolmogorov	Blossom VI	Reported faster than Blossom V on the hardest instances tested; not tried here

Here N is the largest integer cost. Two lessons follow for the CPP. First, the worst-case bounds have improved far more than road networks need: with the sparse pricing of section 4.4, a well-engineered code is limited by the memory for distances, not by the matching. Second, the most active work now comes from quantum error correction, where decoders must pair millions of defects per second. Their local, sparse search is the natural next step for very large postman instances.

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5. Eulerian circuits

Once every degree is even, Hierholzer's algorithm (1873) writes out the tour in time proportional to the number of edges. It is the algorithm the 1997 project already sketched in its section 5.2.2, now implemented so that each edge is looked at once.

The idea follows the constructive proof of Euler's theorem. Walk from the start vertex along unused edges until you are stuck; with all degrees even, you can only get stuck back at the start, so you have a closed sub-tour. Then return along that sub-tour to the first vertex that still has unused edges, walk a new sub-tour from there and splice it in. Repeat until no unused edges remain.

The iterative version uses a stack:

1. Push the start vertex.
2. Look at the vertex v on top of the stack. Advance v's pointer past any edges already used.
3. If v has an unused edge (v, w), mark it used and push w.
4. Otherwise pop v and append it to the output.
5. When the stack is empty, the output reversed is the Eulerian circuit.

Each vertex keeps a pointer into its adjacency list that only moves forward, and each edge is marked used once. The total work is therefore O(n + m). This per-vertex pointer is the refinement that makes the 2026 implementation strictly linear in the size of the graph.

Fleury's rule, discussed in the 1997 report, also works but must test at each step whether an edge is a bridge of the remaining graph. Even with modern dynamic connectivity structures this is slower and more complicated than Hierholzer, so it is no longer used in practice.

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6. Implementation

The solver is a small Python 3 package, `chinese_postman`, built on NetworkX and covered by 425 automated tests (366 when the optional LEMON library is not built). It replaces the six Turbo C modules of 1997 (about 1,000 lines plus custom tree, list and matrix structures) with eleven modules of plain Python and an optional compiled matching library. A JavaScript port of the same solver runs the interactive website (section 6.6).

6.1 Modules

Module	Replaces (1997)	Responsibility
`io.py`	`GRAFO.C`, `leer_archivo`	Reads and writes the original `.GRA` format; reads the CARP `.dat` benchmark format
`cpp.py`	`MAIN.C`, `FULKER.C`, `ARBOL.C`, `SIMETR.C`	Undirected CPP (Edmonds' method) and directed CPP (minimum-cost flow)
`fast.py`	— (new)	Large networks: SciPy Dijkstra and LEMON with dual pricing and a certificate, called through a small C wrapper in native/
`windy.py`	— (new)	Windy postman: bounds, branch-and-cut, Win's method, local search and combined method
`mixed.py`	— (new)	Mixed CPP: bounds, exact branch-and-cut (HiGHS via SciPy), heuristic with local search, mixed instance generators
`bnb.py`	`ARBOL.C`, `MAIN.C`	The branch-and-bound of section 8.3, kept to measure the search; the solvers do not use it
`euler.py`	`EULERIAN.C`	Hierholzer's algorithm on multigraphs, directed or not
`instances.py`	— (new)	Generators for the test families of chapter 7
`web.py`	— (new)	Builds networks for the website from examples, .GRA and CARP files and OpenStreetMap; reference for the JavaScript port

Module	Replaces (1997)	Responsibility
`rural.py`	— (new)	Service and travel costs; rural postman: Frederickson's heuristic, lower bounds, branch-and-cut, combined solve_rural (chapter 10)
`fleet.py`	— (new)	Several postmen (optimal split, local search, bounds, exact model for small networks) and capacitated fleets
`sequence.py`	— (new)	Routes as sequences of served streets: cost, best directions, insertion, 2-opt

Graphs are NetworkX multigraphs, so parallel streets between the same two junctions are kept, as the 1997 adjacency lists allowed. When computing shortest paths only the cheapest of any parallel edges matters, and the solver works on that simple graph internally.

6.2 Pipeline

1. Check that the graph is connected and collect the set T of odd-degree vertices.
2. Run Dijkstra from each vertex of T, keeping distances and paths to the other vertices of T: O(|T| (n + m) log n) with NetworkX's binary heap.
3. Build the complete graph on T and call `min_weight_matching`.
4. Add one copy of every edge on the shortest path of each matched pair.
5. Run Hierholzer on the augmented multigraph and return the tour.

The result object holds the original, added and total costs, the duplicated edges, the matched pairs, the tour as a sequence of vertices, and the time spent in each phase.

6.3 Directed graphs

For a directed graph the imbalance of each vertex, out-degree minus in-degree, plays the role of odd degree. Vertices with more arcs in than out must send extra walks to vertices with more arcs out than in. Choosing those walks at least cost is a transportation problem, solved with NetworkX's `min_cost_flow` on the street network itself, with unbounded arc capacities. The flow on each arc is the number of extra copies to add. The graph must be strongly connected. This reduction goes back to Edmonds and Johnson (1973).

6.4 Testing

Test group	Cases	What it checks
1997 example	1	Added cost 9, total 63, matching {(1,4), (3,6)}

Test group	Cases	What it checks
Classic graphs	3	Königsberg, Petersen and dodecahedron give the known optimum
Brute force	30	Random graphs with up to 8 odd vertices: same cost as enumerating every pairing
Benchmark bound	23	On gdb, where every edge is required, the CPP never exceeds the published CARP lower bound
Structure	5	Parallel edges, generated cities and directed grids give valid closed tours
LEMON backend	59	Same optimum as NetworkX on 51 benchmark files, on towns where a one-neighbour start forces several pricing rounds, and with the certificate checked
Windy postman	51	Exact equals enumeration on 12 small networks; equal costs give the undirected CPP and forbidden directions the mixed CPP; Win is optimal on Eulerian networks and within twice the optimum; measured results agree with published values
Mixed CPP	41	Exact equals brute-force enumeration on 25 random networks; bound ≤ optimum ≤ heuristic; all-two-way and all-one-way cases equal the pure CPPs; the combined strategy proves optimality, including through the cutoff
Branch-and-bound	36	Bounds of the 1997 worked example; same optimum as blossom on 30 random cases (10 for each of the three variants); strict node limit
Certificates and instance reader	18	The dual certificate accepts optima, rejects worse matchings, low duals and imperfect matchings, and equals blossom on 10 random cases; published mixed files are read correctly
Web builders	14	Examples, .GRA, CARP and OpenStreetMap input give the expected networks and solve; bad input gives a clear message
JavaScript solver	243 networks	The website solver gives the Python optimum on every undirected and directed case, never claims a false optimum on mixed ones, and returns a valid tour
Rural postman	36	Exact equals enumeration of every multiset of extra passes on the original network; Frederickson within 3/2 of the optimum; every street served; all-required equals the CPP; first-pass costs, undirected and directed
Rural postman, every network	45	Exact equals enumeration on small windy, mixed and directed networks; bound ≤ exact ≤ heuristic; all-required cases equal the windy postman; measured results agree with their bounds and with published optima
Several postmen	32	Bound ≤ exact optimum ≤ heuristic; split within 2 − 1/k of the optimum; split equals enumeration of cut points; one postman equals the rural postman; capacities respected; on windy, mixed and directed networks; no measured route falls below a published bound
JavaScript extensions	91 networks	Website rural postman and several postmen: valid routes, every street served, never below the Python optimum, bounds never above it

Every test that produces a tour also verifies it: it is closed, it covers every original edge, and its length equals the original edges plus the duplicates.

6.5 Usage

```
from chinese_postman import read_gra, solve_undirected

sol = solve_undirected(read_gra("examples/example_1997.gra"), start=1)
print(sol.summary())
```

6.6 Web version

The project website, miquelmp.github.io/chinese-postman-problem, lets anyone solve a postman problem in the browser: one of the built-in examples or benchmark networks, a `.GRA` file, or a district chosen on an OpenStreetMap map. For an immediate answer the solver is ported to JavaScript and runs inside the page, with no server and nothing to install.

Part	File	Content
Matching	`site/assets/cpp.js`	Edmonds' blossom algorithm, exact, following Van Rantwijk's reference implementation
Directed and mixed streets	`site/assets/cpp.js`	Minimum-cost flow; lower bounds and the mixed heuristic of chapter 9, with its starting orientations, balancing and local search
Tour	`site/assets/cpp.js`	Hierholzer's algorithm, as in chapter 5
Networks	`site/assets/networks.js`	Built-in examples, `.GRA` and CARP readers, OpenStreetMap import through the Overpass API
Extensions	`site/assets/extensions.js`	Rural postman and several postmen on every network (chapter 10): Frederickson's heuristic, local search, optimal split into routes, lower bounds

The port is checked against the Python package. On 243 networks, from classic graphs to benchmark files and mixed towns, it returns the same optimum in every undirected and directed case, stays a mean 0.25% above the optimum on mixed ones, and always returns a valid closed tour; on 3,000 random matchings it agreed with NetworkX every time. The page is ready to solve in under a second.

Only one step still needs Python: proving that a mixed tour is optimal. On request the page loads Pyodide, Python compiled to WebAssembly, with SciPy and the `chinese_postman` package, and runs the branch-and-cut of chapter 9 inside the browser in a few seconds.

The interactive version accepts files of up to 2,500 junctions and OpenStreetMap areas of up to about one square kilometre, or 1,500 junctions. Larger networks are better solved with the Python package and its LEMON backend. The page's service settings choose which streets need service (every street, a random share, the main roads of an OpenStreetMap area, or those marked in a benchmark file) and how many postmen share them, with the methods of chapter 10.

6.7 Numerical tolerances and stopping rules

The blossom code works in integer arithmetic on integer costs; JavaScript numbers are exact below 2^53. The local searches accept a move only if it saves more than 1e-9.

LEMON keeps its duals as integers scaled by 4, and the certificate of section 4.4 keeps vertex prices doubled, so half-integer prices stay exact. A pair fails if its slack is negative, with no margin, and the matching and the dual must agree within 1e-6, far below the cost unit of 1.

HiGHS works in floating point. Its optimality statements are therefore numerical, not exact rational proofs. Every run sets the relative gap to 0 and leaves the other tolerances at their defaults in SciPy 1.18.1: absolute gap 1e-6, integer feasibility 1e-6, primal and dual feasibility 1e-7. A result counts as proved only when HiGHS reports optimality before its time limit or, with a cutoff, reports that nothing beats the heuristic tour. With integer costs a lower bound is rounded up after subtracting 1e-6, and a heuristic tour equal to that bound is optimal.

6.8 Related software

Two open packages and one website cover part of the same ground. OAR Lib (Lum, Golden and Wasil, 2019), in Java under the MIT licence, is the broadest: exact methods for the undirected and directed CPP and heuristics for the mixed, windy, rural and multi-vehicle variants. Its last commit dates from February 2019. `postman_problems` (Brooks), in Python, solves the CPP and the rural postman problem with connected required streets; its last release dates from 2018. This package adds exact methods for the mixed, windy and rural cases, certified matching at large scale and a browser version.

Route Crafter (2025–), under the MIT licence, is the nearest website. It runs OAR Lib's solvers in the browser on OpenStreetMap data and produces a GPX route, for collecting street-level imagery. It is a practical tool and does not report proofs of optimality or measured gaps, which are the focus here.

7. Computational experiments

Every instance tested, 92 distinct benchmark networks (53 of them from real road data) and synthetic cities with up to 40,000 junctions, was solved to proven optimality, including one 40,000-junction town that needed the memory-saving route of section 7.4. None comes from the 1997 Vilanova i la Geltrú case studies. All runs used one core of a 2026 cloud Linux container, Python 3 and NetworkX 3.7; `results/environment.json` records the machine of the latest run (Intel Xeon at 2.10 GHz, 2 cores, 7.8 GB). The timings come from separate runs, so they are comparable within a table and not across chapters.

7.1 Instance families

Family	Source	Networks	Junctions	Streets	Odd vertices
Classic graphs	Königsberg, Petersen, dodecahedron, 1997 example	4	4–20	7–30	4–20
gdb	Golden, DeArmon and Baker (1983), random networks	23	7–27	11–55	0–16
val	Benavent et al. (1992), generated networks	10	24–50	34–97	12–28
kshs	Kiuchi et al. (1995)	6	6–10	15	4–6
egl	Li and Eglese (1996), Brandão and Eglese (2008): winter-gritting roads in Lancashire, UK	3	77–255	98–375	50–190
Beullens C, E (D, F share their networks)	Beullens et al. (2003): road network of Flanders, Belgium	50	26–97	35–142	14–54
Grid city	Square grid, short blocks and long avenues, 10% of segments closed	12	100–40,000	161–71,752	42–12,066
Delaunay town	Random junctions at a constant density (a square of side 100 m × √n), Delaunay triangulation thinned to 75%, lengths of at least 1 m	12	100–40,000	214–89,977	48–19,942
One-way grid	Directed grid, alternating one-way streets, two-way outer ring	5	48–3,600	113–8,142 arcs	18–204 unbalanced

Benchmark files that share a road network and differ only in demands (for example val1A, val1B and val1C) were solved once, leaving 92 distinct networks out of 254 files. Families A

and B in the same repository, of unknown origin, were left out. Delaunay towns keep a constant density of junctions: they are drawn in a square of side 100 m × √n, so streets average about 88 m on the largest towns. Lengths are rounded to whole metres, with a minimum of 1 m, so no street has a cost of 0.

7.2 Classic graphs

Graph	Odd vertices	Streets	Added	Total	Tour length
1997 example	4	12	9	63	15 edges
Königsberg bridges	4	7	2	9	9 crossings
Petersen graph	10	15	5	20	20 edges
Dodecahedron	20	30	10	40	40 edges

Euler's citizens of Königsberg could not cross each bridge exactly once, but a postman can cover all seven with nine crossings by repeating two. In the Petersen graph and the dodecahedron every vertex is odd, and both contain a perfect matching of unit edges, so the tour repeats one edge for every two vertices: a third of the edges.

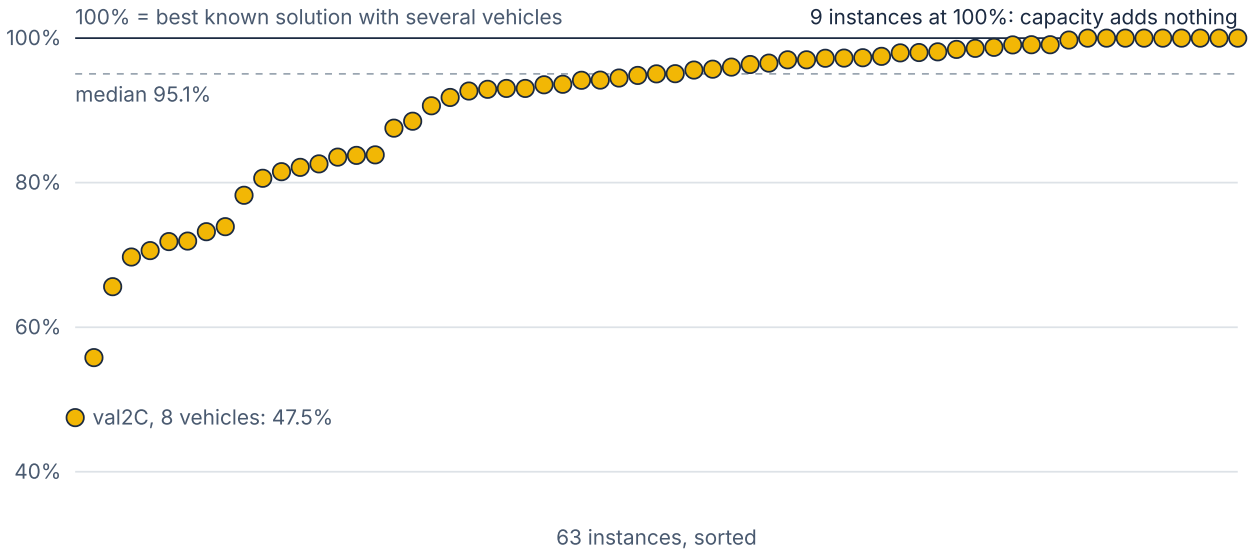
7.3 Benchmark networks

Family	Networks	Added distance, % of network (mean)	Range	Slowest solve
gdb	23	8.9%	0–22.2%	0.003 s
val	10	14.9%	10.9–18.5%	0.007 s
kshs	6	14.9%	10.5–20.4%	0.001 s
Beullens C	25	20.1%	11.4–24.3%	0.06 s
Beullens E	25	19.4%	12.3–26.9%	0.06 s
egl-e	1	37.4%	—	0.04 s
egl-s	1	24.5%	—	0.21 s
egl-g	1	24.4%	—	2.46 s

On real road networks a postman must repeat about a fifth of the distance; on the sparse Lancashire network of egl-e, more than a third. The largest benchmark, egl-g, with 255 junctions and 190 odd vertices, takes 2.5 seconds, with optimality proven.

In 39 of these networks every street needs service, so any multi-vehicle CARP solution is also a postman tour, and the CPP optimum is a lower bound for it. Files that share a road network were solved once above, but they differ in capacity and number of vehicles, so the comparison with fleets uses all 63 gdb, val and kshs files in which every street is required. The solver respects the bound on all 63. How close it comes depends on capacity:

One postman costs a median 95.1% of the best known fleet, and as little as 47.5%



results/cpp_bound.csv · 63 all-required instances from gdb, val and kshs; best known CARP solutions as published in CARPData.jl

Instances	Number	Median	Lowest
gdb	23	96.0%	71.8%
kshs	6	88.3%	65.6%
val A (fewest vehicles)	10	99.1%	95.1%
val B	10	96.0%	83.8%
val C	10	78.7%	47.5%
val D (four networks only)	4	76.9%	71.9%
All	63	95.1%	47.5%

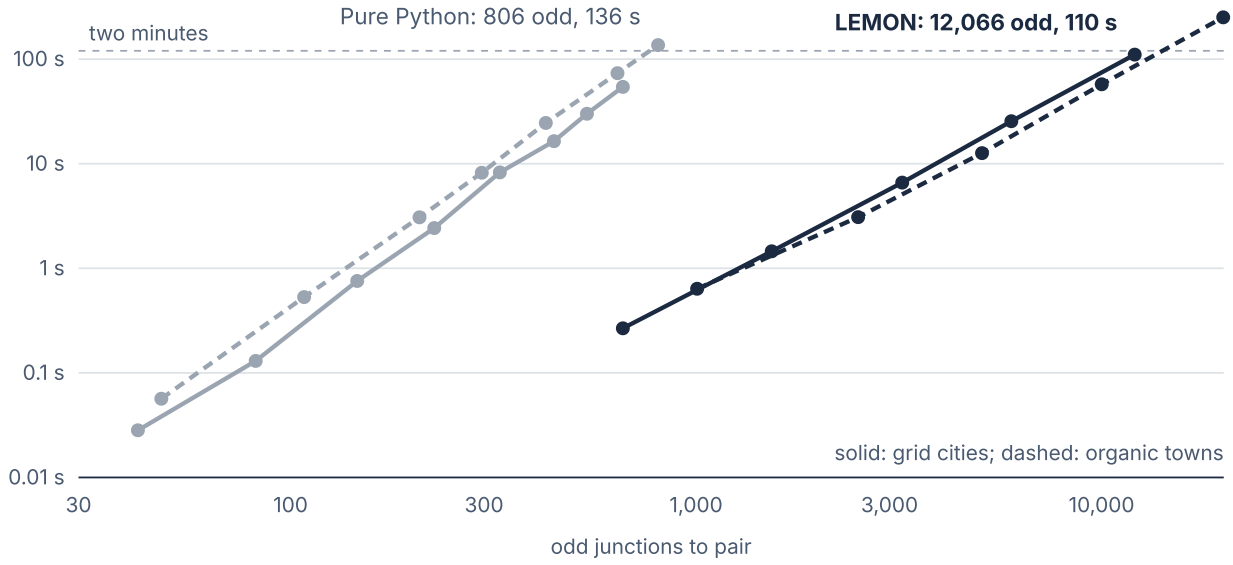
The mean is 90.3%; 19 instances fall below 90% and 10 below 80%, while 9 reach 100%, where capacity adds nothing to the postman's tour. On each of the 10 val networks the bound weakens steadily as capacity shrinks and vehicles are added: val2 goes from 95.6% with 2 vehicles to 47.5% with 8, because every vehicle must drive out from and back to the depot. Vehicle count

does not explain everything: the lowest kshs and gdb values are kshs1 (65.6%, 4 vehicles) and gdb8 (71.8%, 10 vehicles), while gdb23 stays at 95.7% with 10 vehicles. In all 63 files the published lower and upper bounds coincide, so these fleet costs are proven optima.

7.4 Scaling

With the LEMON backend, a network with 40,000 junctions and 12,066 odd vertices solves to proven optimality in a median 110 seconds, against about 750 odd vertices in the same time with NetworkX. These two figures come from different families; within one family, LEMON reaches 12,066 odd vertices in 110 s on grids, against 660 in 54 s with NetworkX, and 10,006 in 57 s on Delaunay towns, against 806 in 136 s. Grid cities and Delaunay towns were generated with up to 40,000 junctions; both backends were run, NetworkX until a case passed 90 seconds.

With LEMON, two minutes now covers 12,000 odd junctions instead of about 750



results/timing_repeats.csv: medians of five runs (NetworkX, 15 networks; LEMON, 10 networks) · 100–40,000 junctions

Network	Junctions	Odd vertices	Shortest paths	Matching	Tour	Total, median	Total, range	Pricing rounds	Peak memory
Grid city	2,025	660	0.13 s	0.04 s	0.10 s	0.27 s	0.2–0.3 s	1	0.1 GB
Grid city	10,000	3,224	4.0 s	0.42 s	2.1 s	6.6 s	5.5–6.7 s	2	0.4 GB

Network	Junctions	Odd vertices	Shortest paths	Matching	Tour	Total, median	Total, range	Pricing rounds	Peak memory
Delaunay town	10,000	5,072	7.7 s	1.1 s	4.0 s	12.6 s	12.5–13.6 s	2	not measured separately
Grid city	19,600	5,988	15.1 s	2.5 s	7.7 s	25.5 s	22.1–25.9 s	2	0.9 GB
Delaunay town	20,000	10,006	34.5 s	5.1 s	17.9 s	57.3 s	56.3–59.6 s	2	not measured separately
Grid city	40,000	12,066	67.2 s	8.5 s	33.9 s	110 s	99–111 s	3	2.0 GB
Delaunay town	40,000	19,942	150 s	25.1 s	75.7 s	250 s	249–261 s	3	2.8 GB

Each network was run five times, in a cycle over all the networks, with nothing else on the machine (`results/timing_repeats.csv`). The table gives medians. The matching column includes pricing and the certificate. Every repeat gave the same optimum. For networks taking a second or more, the range is typically 14% of the median. The grids and the towns were timed on consecutive days, on the same kind of machine, and the towns were rerun after the generator changed. Peak memory comes from a single run.

On the 660-odd-vertex grid the fast backend is about 200 times faster than NetworkX, and the matching itself takes 0.04 s instead of 50 s. Starting from 10 nearest neighbours, one to three rounds were always enough, the last of them only confirming optimality; nine of the ten networks needed extra pairs, at most 83. The matching is no longer the bottleneck on large networks: shortest paths take about 60% of the time and building the tour about 30%, both still written in Python around compiled kernels and both easy to speed up further. Memory is dominated by the table of distances between odd vertices, which grows with their square, so networks much bigger than these would mean never storing that table, as the sparse blossom method does.

The certificates of section 4.4 confirm these results a second way. `tools/certify_matching.py` repeats all ten networks with the JavaScript blossom code instead of LEMON, and gives exactly the same optimum on each (`results/certificates.csv`). It is slower, 117 s of matching against 9 s on the largest grid and 200 s against 25 s on the largest town, but two unrelated implementations agree.

For the NetworkX backend, time grows roughly with the cube of the number of odd vertices, as expected from the $O(n^3)$ matching, which takes over 85% of the time above 200 odd vertices.

The repeated distance settles at 12–15% of the network on both families, lower than on the benchmark networks because grids and triangulations are denser.

7.5 Directed networks

Grid	Junctions	Arcs	Unbalanced junctions	Added distance	Solve time
6 × 8	48	113	18	11.7%	0.003 s
10 × 12	120	280	32	10.4%	0.006 s
20 × 20	400	912	64	8.4%	0.02 s
40 × 40	1,600	3,627	134	6.9%	0.14 s
60 × 60	3,600	8,142	204	6.6%	0.39 s

The directed CPP is much faster than the undirected one at the same size because minimum-cost flow works on the sparse street network itself rather than on a complete graph of odd vertices. In a city of alternating one-way streets, a postman repeats only 7–12% of the distance.

8. From branch-and-bound to blossom: improvements in this version

The 2026 version keeps the structure of the 1997 method and improves each of its stages. The largest gain comes from solving the matching exactly in polynomial time; a tightened lower bound also makes the branch-and-bound itself much faster, which matters for the harder variants of chapter 9 where B&B remains the method of choice.

8.1 Summary of improvements

Stage	1997 version	2026 version	Effect
Matching	Binary branch-and-bound with a time limit for large cases	Edmonds' blossom algorithm	Proven optimum at every size, polynomial time

Stage	1997 version	2026 version	Effect
B&B lower bound	Row minima divided by half the number of odd vertices	Row minima divided by 2	Far fewer nodes; useful for B&B on harder variants
Shortest paths	Bellman-type iteration on the full V × V matrix	Dijkstra from odd vertices only	Work proportional to the odd vertices, not to all pairs
Eulerian circuit	Cycle splicing over adjacency queues	Hierholzer with one pointer per vertex	Linear time
Directed streets	Future work	Minimum-cost flow	Polynomial (Edmonds and Johnson, 1973)
Platform	Turbo C, 16-bit integers, 640 KB memory model	Python 3, unbounded integers, automatic memory	No size limits on costs or data
Verification	Manual checks	425 automated tests in Python and 334 networks and 3,000 matchings re-checked in JavaScript, every tour validated	Results checked on each run

8.2 The 1997 branch-and-bound

The choice of branch-and-bound in 1997 was deliberate. The project was an academic exercise. Its report presents and proves Edmonds and Johnson's algorithm in chapter 3, and then proposes "a method less specific and conceptually simpler": to treat the pairing as a general combinatorial problem. As its author recalls, the practical interest lay in large cases, where the search reached fairly good solutions quickly, while the exact method took much longer on the computers of the time.

Branch-and-bound is itself exact when it runs to completion, and it becomes a heuristic only when a time or memory limit stops it, still with a bound on its error. Its cost can grow exponentially where the blossom algorithm is polynomial, but it needs no matching theory, and it carries over to problems with no polynomial method, such as the mixed CPP of chapter 9.

Each node of the search tree fixed some pairs of odd vertices as taken and others as rejected. Branching took the cheapest pair still available and created two children: take it or reject it. The lower bound of a node was $c_1 + c_2$, where c_2 is the cost of the pairs already taken and c_1 estimates the cost of completing the matching from the cheapest available pair at each remaining vertex. Nodes were explored lowest bound first while the tree was small, and deepest first afterwards, which gives good solutions early and makes the method a natural time-limited heuristic.

8.3 A tightened lower bound

Every matched pair is reached from both of its ends, so the cost of any perfect matching M is half the sum, over all odd vertices, of the pair each one belongs to. That gives the bound

$$c(M) = \tfrac{1}{2}\sum_{v \in T} d\big(v, M(v)\big) \;\geq\; \tfrac{1}{2}\sum_{v \in T} \min_{u \neq v} d(u, v)$$

The 1997 version divided the same sum by half the number of odd vertices, imp/2. The two coincide when there are four odd vertices, as in the worked example; with more odd vertices the divisor 2 gives a higher, tighter bound and prunes much more of the tree.

Both bounds were measured with a reimplementation of the 1997 search, `chinese_postman/bnb.py`, which follows the Turbo C listings and reproduces the bounds of the 1997 worked example. Only the divisor changes between the two runs. The instances are five grid towns for each number of odd vertices, and the limit is 200,000 explored nodes, so the counts do not depend on the machine:

Odd vertices	1997 bound: proven	Nodes (median)	Tightened bound: proven	Nodes (median)	Nodes (worst)
8	5 of 5	41	5 of 5	6	9
12	5 of 5	1,925	5 of 5	18	55
16	4 of 5	68,058	5 of 5	54	349
20	0 of 5	limit reached	5 of 5	53	1,569
24	0 of 5	limit reached	5 of 5	141	1,052
30	0 of 5	limit reached	5 of 5	1,656	44,740

Even when stopped at the limit, the 1997 search had usually found the optimum: in 14 of its 16 stopped runs, and within 7% in the other two. The counts vary widely between towns of the same size, so they show orders of magnitude and not a precise law.

Two further changes extend the reach: branching on the most constrained vertex, and starting from a greedy pairing improved by pair swaps. With both, the search proves all five towns at 40 and at 50 odd vertices, in under 10 seconds each, and four of five at 60; the tightened bound alone proves four of five at 40 and three of five at 50. They do not help everywhere: at 30 odd vertices the worst case rises from 44,740 to 106,867 nodes. Blossom solves the same cases of up

to 50 odd vertices in under 0.1 s, which is why the 2026 version uses it for the CPP and keeps the B&B for problems with no polynomial algorithm.

8.4 Why the B&B still matters

The 1997 conclusion, that a time-limited B&B yields very good solutions quickly on large instances, still holds and carries over to NP-hard arc routing problems such as the mixed CPP and the rural postman problem. For those, the improvements above (a tight bound, a good first solution and a branching rule that fixes the hardest decision first) are exactly the ingredients modern exact methods rely on.

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9. The mixed Chinese Postman Problem

Real towns mix one-way and two-way streets. This mixed CPP is NP-hard (Papadimitriou, 1976), so no polynomial algorithm is known; the 2026 version solves it exactly by branch-and-cut and approximately by a fast heuristic whose distance from the optimum is measured.

9.1 Why it is harder

The two pure cases are easy for different reasons. An undirected network needs only even degrees, fixed by a matching (chapter 3); a directed one needs only as many arcs in as out at each junction, fixed by a minimum-cost flow (chapter 6). In a mixed network the direction in which each two-way street is driven and the choice of streets to repeat must be decided together, because each decision changes the other's constraints.

A mixed network has a postman tour only if it is strongly connected when two-way streets count in both directions. The generators below make sure of it.

9.2 Model and exact method

Let x(u → v) be the number of times the tour drives a street from u to v:

$$\min \sum c\,x \quad \text{s.t.} \quad \sum_{\text{out of } v} x = \sum_{\text{into } v} x \;\; \forall v, \qquad x(u \to v) \geq 1 \text{ for a one-way street}, \qquad x(u \to v) + x(v \to u) \geq$$

Any integer solution is a balanced, connected multigraph, so it contains an Eulerian circuit: the tour. The integer program is solved with HiGHS, an open-source solver bundled with SciPy,

with its relative gap tolerance set to zero (the default of 0.01% would stop short of a proof). HiGHS runs branch-and-bound with cutting planes: the 1997 idea of separating and bounding, with the bounds supplied by linear programming.

9.3 Bounds and heuristic

Two lower bounds are cheap to compute. The linear relaxation of the model ignores parity and lets a two-way street be driven half each way; the undirected CPP of the network ignores directions. Each misses what the other captures, so the larger of the two is used.

The heuristic is polynomial and runs in four steps:

1. Build four starting orientations of the two-way streets: two roundings of the linear relaxation, and the direction in which an Eulerian circuit of the undirected CPP solution drives each street, read forwards and backwards.
2. For each orientation, add the cheapest extra traversals that balance every junction. With directions fixed this is a minimum-cost flow, solved exactly.
3. Improve the two best by local search. The flow's dual prices m estimate what reversing a street $a \to b$ would change, $2 \cdot (m_b - m_a)$; the most promising reversals are tried, each verified by an exact re-solve, and kept only if they lower the cost.
4. Return the best tour found, with the best lower bound as a certificate of how far it can be from the optimum.

The orientation-then-balance idea goes back to Frederickson (1979). The local search is what makes it accurate: across the 121 benchmark and town instances of section 9.4 with a proven optimum, it improved the best starting orientation on 77 and cut the mean gap to the optimum from 1.66% to 0.41% and the worst from 11.9% to 4.8%. Most of the gain is on the benchmarks (mean 1.58% to 0.12%); on the 29 towns the mean falls only from 1.92% to 1.35%.

9.4 Results

On the benchmark networks and on towns of up to 400 junctions the mixed CPP is solved exactly in seconds. Above that, size alone does not decide: some towns are proved in under a minute and others of the same size are not proved in five. The heuristic is 1% to 3.6% above the best tour of branch-and-cut on the towns left unproved, in a fraction of its time; its worst gap to a proven optimum is 4.8%. On a published test set of 500 to 3,000 junctions the heuristic is weaker where few streets are one-way, up to 5.8% above the published optimum, and much slower.

Benchmark networks. In each of the 92 distinct networks of chapter 7, each street was made one-way in a random direction with probability 0.4. Streets that broke strong connectivity went back to two-way, leaving 31% of all streets one-way (24–44% per network).

Result	Value
Networks solved to proven optimality	92 of 92
Slowest exact solve	5.8 s (egl-g, 255 junctions, 375 streets)
Heuristic finds the optimum	78 of 92
Heuristic gap to optimum	mean 0.12%, maximum 2.8%
Slowest heuristic run	3.8 s

Mixed towns. Grid towns in which each whole street is one-way with probability 0.6, as in a typical town centre, from 64 to 6,400 junctions; five towns per size, a 5-minute limit for the exact method.

Junctions	Streets	Proved in 5 min	Exact: time when proved	Heuristic: time	Heuristic: gap to the optimum, when proved	Unproved: exact to its bound	Unproved: heuristic above the exact tour
64	112	5 of 5	under 2 s	0.1–0.4 s	0–2.6%	—	—
144	264	5 of 5	4.1–5.2 s	0.3–1.1 s	0.2–2.8%	—	—
256	480	5 of 5	2.4–7.5 s	0.4–1.0 s	0.3–2.8%	—	—
400	760	5 of 5	1.6–5.5 s	0.6–2.8 s	0.2–1.8%	—	—
900	1,740	4 of 5	9–190 s	3.4–8.4 s	0.7–3.3%	0.47%	3.6%
1,600	3,120	4 of 5	16–97 s	7–22 s	1.6–4.8%	0.14%	2.0%
3,600	7,080	1 of 5	38 s	25–51 s	2.3%	0.48–1.01%	1.0–2.4%
6,400	12,640	0 of 5	—	64–98 s	—	0.25–0.97%	1.0–2.0%

Published instances. The benchmark networks and towns above were made mixed by this project, so they may favour its heuristic. As an outside test, the 120 mixed instances of Corberán, Oswald, Plana, Reinelt and Sanchis (2012) were read unchanged, in the copy distributed with OAR Lib (Lum, Golden and Wasil, 2019). They have 500 to 3,000 junctions; half are random graphs (MA) and half imitate street networks (MB). The authors proved 108 of them with a dedicated branch-and-cut and a 10-hour limit, and their instance page now lists 109 optima, with lower and upper bounds for the other 11.

Two sets were run, each with a 2-minute limit for the exact method. A sample of 30, every fourth file, covers all sizes. In that sample, however, street density and the one-way share change together, so all 48 instances of 500 and 1,000 junctions were also run. Twelve instances are in both sets, which makes 66 distinct instances. The two sets shared a two-core machine for much of the run, so times may be high and some proofs may have missed the limit.

Junctions	Instances in the sample	Proved in 2 min	Heuristic: time	Unproved: exact to its bound
500	6	6 of 6	8–13 s	—
1,000	6	4 of 6	28–70 s	0.18–0.21%
1,500	6	4 of 6	63–213 s	0.10–0.39%
2,000	6	1 of 6	154–483 s	0.03–0.90%
3,000	6	2 of 6	443–1,528 s	0.15–0.72%

Agreement with the published values. The 39 distinct instances proved here all have the published optimum (`results/mcpp_published_values.csv`). On all 66, no tour falls below the best published value, no bound exceeds it, and the exact tour is at most 0.72% above it. This is an external check of the model, of the file reader and of the solver. The times are not comparable with the published ones, which come from another solver and a 2007 machine.

Heuristic against the published optima. On the 29 sample instances with a published optimum, the heuristic averages 2.09% above it, with a worst case of 5.79%. Across all 48 instances of the full set the mean is 2.03% and the worst case 5.33%. Each cell below is the mean of four instances, one for each combination of size and family.

Street density (middle digit of the name)	About 25% one-way	About 50% one-way	About 75% one-way	Mean
3	4.29%	1.42%	0.27%	1.99%
4	4.43%	1.38%	0.27%	2.02%
5	4.25%	1.85%	0.15%	2.08%
6	3.60%	2.17%	0.29%	2.02%
Mean	4.14%	1.70%	0.25%	2.03%

The three columns have 21–26%, 42–51% and 57–76% of streets one-way. In this set the exact method proved 32 of the 48 in two minutes.

Real street networks. Five OpenStreetMap extracts, taken on 3 November 2026, were solved with their one-way tags as they stand, with a 5-minute limit for the exact method (`data/osm`, `results/mixed_osm.csv`). The areas were chosen by hand to span the one-way share, Somerville among them to fill the middle of the range, so they are examples and not a sample of cities. Each network keeps its largest strongly connected part, which drops 5–11% of the street segments.

Area	Junctions	Streets	One-way	Length	Exact: optimum, time	Heuristic: time	Heuristic above the optimum
Eixample area, Barcelona	1,154	1,699	92.9%	114.3 km	190.6 km, 1.5 s	22 s	0.00%
Lower Manhattan, New York	1,111	1,633	82.2%	111.4 km	158.2 km, 2.7 s	10 s	0.08%
Vilanova i la Geltrú	1,065	1,468	75.5%	79.9 km	132.4 km, 8.1 s	56 s	1.06%
Somerville, Massachusetts	1,104	1,443	36.9%	108.4 km	153.7 km, 34 s	116 s	3.07%
Freehold, New Jersey	341	472	10.8%	70.7 km	100.8 km, 3.0 s	6 s	1.65%

Branch-and-cut proves all five in at most 34 seconds. A tour drives 42–67% more than the length of the streets, far above the 12–15% of two-way networks in section 7.4. The heuristic's gap generally grows as the one-way share falls, as on the published set, though Freehold, much the smallest, breaks the pattern and Vilanova is above the 0.25% of published instances with a similar share. Five networks show that the results carry over to real streets, but they cannot fix a trend.

Five conclusions follow.

1. **Exact is practical up to a district, but not always.** Branch-and-cut proves every benchmark network and all 20 towns of up to 400 junctions within 8 seconds, 8 of the 10 towns of 900 and 1,600 junctions, one of five at 3,600 and none at 6,400. Every unproved town still ends within about 1% of its proven bound in five minutes, with a certificate.
2. **The heuristic is fast, and more accurate on the benchmarks than on towns.** With a proven optimum it averages 0.12% above it on the 92 benchmarks (worst 2.8%) and 1.35% on the 29 towns (worst 4.8%). Its worst cases are towns where single-street reversals get stuck; moves that reverse whole cycles of two-way streets would be the next improvement.

3. **Bounds matter.** The cheap combined bound averages 94.8% of the optimum on the benchmarks, as low as 83%, so on its own it certifies little. The bound computed by branch-and-cut is far tighter: it is what turns a good solution into a certified one.
4. **The heuristic's accuracy depends on the one-way share.** On the published set its mean gap falls from 4.14% with about a quarter of streets one-way to 0.25% with about three quarters, while street density makes almost no difference (1.99–2.08%). Its running time also grows much faster than on the towns: 7 to 25 minutes at 3,000 junctions, against under 100 seconds at 6,400 in a town. From 1,500 junctions upwards, branch-and-cut gave a better tour than the heuristic on all 18 sample instances, in less time on 17. The cause of the slowdown was not isolated.
5. **Real streets are easier to prove.** The five real networks, of up to 1,154 junctions, are proved in at most 34 seconds, while only half of the published instances of 1,000 junctions are proved in two minutes. The heuristic stays within 3.1% of the optimum on them.

9.5 Combined strategy

In practice the two methods are used together, through a single function, `solve_mixed`, that returns the best tour found and a proven lower bound:

1. Compute the cheap bounds and run the heuristic, which gives a good tour in seconds.
2. If the heuristic already matches the bound, or is within a chosen tolerance of it, stop.
3. Otherwise run branch-and-cut for the remaining time, asking it only for tours cheaper than the heuristic's. If none exists the model is infeasible, which proves the heuristic tour optimal; if one exists, it replaces the heuristic tour.

The objective cutoff lets branch-and-cut discard every branch that cannot beat the heuristic, so the heuristic's quality speeds up the exact search. On a 64-junction town where the heuristic was already optimal, this proves it in under a second. On a 1,600-junction town where the heuristic alone was 4.8% above the optimum, the combination returns the optimum in 79 seconds. With a tolerance of 0.5%, towns of up to 256 junctions stop in about three seconds or less with a tour certified within that margin.

This extends the approach of chapter 12 of the 1997 report, a time-limited exact search used as a heuristic: there is now a branch-and-cut solver behind it, and a proven bound on the remaining error.

10. The paths left open in 1997

The 1997 report closed with two proposed extensions, the asymmetric case and a cost that depends on the pass, and with an observation that a working shift limits the work. All three are followed up here, with code, tests and measurements. This chapter follows them in turn, and for each problem states how hard it is, whether it is solved exactly or by a heuristic, and how far the heuristic can be from the optimum.

Starting point in 1997	Where it leads	Chapter
"La resolución de CPP para el caso asimétrico" (the asymmetric case)	Directed and mixed CPP	6.3, 9
"[...] que el valor asociado en cada arista variara en función de que fuera o no la primera vez que se recorre" (a different cost for the first pass)	Service and deadheading costs; the rural postman problem	10.2, 10.3
"[...] el trabajo que se puede realizar en media jornada [...] es claramente limitado" (a shift limits the work)	Several postmen; capacitated fleets	10.4, 10.5

10.1 Exact, heuristic, bound and guarantee

A problem is polynomial when an algorithm is known whose running time grows like a power of the network size; such problems can be solved exactly on city-sized networks. The undirected and directed CPP are polynomial. A problem is NP-hard when it is at least as hard as every problem in NP, the class of problems whose solutions can be checked in polynomial time: a polynomial algorithm for one NP-hard problem would give one for all of them, and none is known. For these there are two complementary methods. An exact method, here branch-and-cut, finds the optimum and proves it, but its running time can grow very fast. A heuristic answers quickly and is paired with a lower bound, a value that no solution can beat; the distance between the heuristic and the bound, the certified gap, is the most the heuristic can be off. Some heuristics also carry an approximation guarantee: a proof that they are never more than a fixed factor above the optimum, on any network that meets its hypotheses. A lower bound certifies how far a particular solution can be from the optimum. An approximation guarantee establishes a worst-case performance ratio for every instance satisfying the method's assumptions. Experimental results, however good, do not establish a general approximation guarantee. The guarantees of sections 10.3 and 10.4 assume undirected networks with non-negative costs and one cost per street, so serving and travelling cost the same. Distances are shortest paths, so the triangle inequality holds by construction. Section 10.4 also needs an optimal single route. Win's

factor in section 10.6 is for the windy postman with every street served. The mixed heuristic of chapter 9, the capacitated method of section 10.5 and the methods of section 10.7 are not claimed to have any fixed factor.

Whether a polynomial algorithm can exist for the NP-hard problems is the P versus NP question, posed formally by Cook (1971) and, independently, by Levin (1973). It is one of the seven Millennium Prize Problems announced by the Clay Mathematics Institute in 2000, each with a prize of one million dollars; of the seven, only the Poincaré conjecture has been solved, and P versus NP remains open.

If P = NP were proved, every NP-hard problem in this chapter, and the mixed CPP of chapter 9, would have an exact polynomial algorithm, and heuristics would no longer be needed in principle. Two caveats apply: the proof might show that such an algorithm exists without showing how to build it, or it might give one whose running time grows with so high a power of the network size that it is useless in practice. Much of today's cryptography, which relies on some problems being hard, would also be at risk. If P ≠ NP, as most researchers expect, there is no shortcut, and the approach proposed in 1997 and followed here is the right one: an exact search as far as it can go, and beyond that a heuristic with a bound.

The postman shows how thin the line is. Edmonds' method solves the Chinese Postman Problem in polynomial time, yet mixing one-way with two-way streets, leaving some streets without service, or adding a second postman is enough to cross it.

10.2 First and later passes

Let every street e carry a service cost s(e), paid on the pass that serves it, and a travel cost c(e), paid on any other pass. The 1997 examples were a road covered in snow and a street whose bins are still full.

When every street needs service, each is served exactly once, so the service costs add up to a constant, and only the extra passes can change. These must make every degree even, as in chapter 3, and they cost c. Hence

$$\text{cost} = \sum_{e \in E} s(e) + \min_{\text{T-join } J} c(J),$$

which is the ordinary CPP augmentation computed on the travel costs. The problem stays polynomial: matching for undirected networks and minimum-cost flow for directed ones

(`rural.solve_two_costs`). For mixed networks it inherits the NP-hardness of chapter 9.

10.3 The rural postman problem

When only a subset R of the streets needs service, the other streets may still be used to travel. This is the rural postman problem (RPP). It is NP-hard (Lenstra and Rinnooy Kan, 1976), but the difficulty is precise. If the required streets, together with the depot, form one connected piece, the argument of section 10.2 still applies: pair the odd vertices of R by shortest paths through the whole network, and the result is optimal in polynomial time. When R falls into several pieces, choosing how to join them is the hard part.

All three methods work on a reduced graph whose vertices are the endpoints of required streets and the depot. Two of them are joined by a deadhead edge whose cost is their shortest-path distance; edges whose shortest path passes through a third required vertex are left out, since they are never needed. With x(e) the number of extra passes over deadhead edge e, the model is

$$\min \sum_{e} c(e)\,x(e) \quad \text{s.t.} \quad x(\delta(v)) \equiv \deg_R(v) \pmod 2 \ \ \forall v, \qquad x(\delta(S)) \geq 2 \ \ \forall S \text{ union of pieces, } S \neq V,$$

- **Heuristic** (`rural.heuristic`). Frederickson (1979): join the pieces by a minimum spanning tree of their distances, then pair the odd vertices by a minimum matching. It is never more than 3/2 times the optimum (Benavent, Campos, Corberán and Mota, 1985). The route is then improved by moving each street to its best position and by reversing stretches of the route (2-opt), working on the sequence of served streets (`sequence.improve`).
- **Lower bound** (`rural.lower_bound`). The best of two: the odd vertices of R must be paired, so the deadheading costs at least a minimum matching of them; and the linear relaxation of the model, with the cut around every piece and further cuts found by minimum cuts of the fractional solution.
- **Exact method** (`rural.solve_exact`). HiGHS solves the integer program; whenever its solution falls apart, the cut around each component is added and the program solved again. This is the model of Ghiani and Laporte (2000), with the cuts added one round at a time.

`rural.solve_rural` combines them as `solve_mixed` does in chapter 9: heuristic and bound first, then branch-and-cut told to beat the heuristic; if it cannot, the heuristic route is proven optimal.

Networks	Count	Streets to serve	Optimum proved	Frederickson above the optimum (mean, worst)	After local search (mean, worst)	Slowest proof
Benchmark files, required streets in one piece	51	28–347	51, by matching	—	—	3.7 s
Benchmark files, several pieces	6	51–159	6	0.81%, 3.23%	0.03%, 0.18%	6.2 s
Grid towns 10×10 and 20×20, 20% or 50% to serve	12	30–387	10	3.73%, 8.33%	1.04%, 4.75%	74 s
Grid towns 30×30, 20% or 50% to serve	6	330–870	0 in two minutes	—	within 6.39% of the bound	—

The benchmark files are those of chapter 7 whose demands mark only some streets for service (Eglese and Beullens et al.); the towns are grids in which each street needs service with probability 0.2 or 0.5, with a two-minute limit. Frederickson's heuristic stays far inside its 3/2 guarantee: its worst case, on a small town with 15 pieces, is 8.3% above the optimum. The local search removes most of that excess, and branch-and-cut then either proves the route optimal or improves it. On the six largest towns, with 48 to 173 pieces, the optimum is not proved in two minutes, and the bound certifies the route within 0.8% to 6.4%.

10.4 Several postmen

When a district is too large for one shift, k postmen share it, all starting and ending at the depot. The min-max k-postmen problem asks for k routes that together serve every required street, with the longest route as short as possible. It is NP-hard even when every street needs service (Frederickson, Hecht and Kim, 1978).

- **Heuristic** (`fleet.solve_k_postmen`). Write out one postman's route as a sequence of served streets, then cut it into k consecutive pieces. A dynamic programme finds the cut points that make the longest route shortest; each piece becomes a route from the depot and back. Several written-out orders of the same route are tried. A local search then moves and swaps streets between the longest route and the others, re-plans each route as a small rural postman problem, and perturbs the best solution a few streets at a time while time remains.
- **Guarantee.** Let L be the length of an optimal single route, and r the largest cost of going from the depot to a required street, serving it and coming back. Any solution has a route at least r long, and the k routes together cost at least L, so the optimum is at least max(r, L/k).

Frederickson, Hecht and Kim (1978) choose cut points on the optimal single route so that the longest route is at most (2 − 1/k) times the optimum. The dynamic programme considers every cut, so its result is at least as good; the guarantee holds whenever the single route is optimal, which is always the case when every street needs service.

- **Lower bound.** max(r, L/k), with L replaced by its lower bound when the single route is not proved optimal.
- **Exact method** (`fleet.solve_k_exact`), for small networks only: one copy of the rural model per postman, an assignment of every required street to one postman, the longest route as the objective, and connectivity cuts added round by round.

Networks	Count	Measure	Best split of one tour	After local search
Small random networks, 2 or 3 postmen	62	Above the exact optimum (mean, worst)	2.91%, 26.6%	0.02%, 1.25%
Small random networks	62	Optimal	—	61 of 62
Benchmark networks, 2 to 10 postmen	312 runs	Above the lower bound (median, worst)	12.0%, 44.6%	6.2%, 37.1%

On the small networks, half of them with every street to serve and half with 60% (two further cases with three postmen are left out, because the exact method did not finish within a minute), the best split of one tour is never more than 27% above the optimum, well inside the guarantee of 50% for two postmen and 67% for three, and the local search reaches the optimum in 61 of the 62 cases. On the benchmark networks the optimum is out of reach and only the lower bound is available, with 5 seconds of local search per run. A further 105 runs with more than 10 postmen are left out: there the farthest street sets the bound, and the longest route is at most 0.7% above it. Much of the remaining distance is the bound's: on the small networks, where the optimum is known, the same bound is a mean 3.0% and up to 17.5% below it. A stronger bound would be the natural next step.

10.5 Vehicles with a capacity

When each street produces a load and each vehicle carries a limited amount, the problem is the capacitated arc routing problem (CARP), the subject of the benchmark files used throughout this report. It is NP-hard (Golden and Wong, 1981). With the same sequence of served streets, Ulusoy's (1985) dynamic programme cuts the route into vehicle loads at least total cost; streets

are then moved between routes for up to 5 seconds, while the total cost falls and the loads fit (`fleet.solve_carp`). The number of vehicles is left free.

On the 197 classical CARP instances (gdb, val, egl, kshs and the four Beullens families), the routes are a median 5.8% above the best solutions published (mean 5.5%, worst 14.8%), and match them on 16 instances; the split alone is a median 8.8% above. 31 of the 197 best solutions are not proven optimal.

The number of vehicles needs care, because the literature has two conventions. Some authors fix the fleet at the number in the file, and others treat that number only as a lower bound (Bode and Irnich, 2012), while metaheuristics usually let the number of routes vary (Chen, Hao and Glover, 2016). This study also leaves it free. The published tables examined do not list the vehicles of each best solution, so the comparison below could not be made solution by solution.

Instances	Count	File's vehicles	Routes against the file	Above the best published (median, mean)
gdb, val, egl, kshs	32	Tight: demand fills 90% of the fleet on average	Within it	3.2%, 3.3%
gdb, val, egl, kshs	65	Tight: demand fills 95%	Up to 4 more	6.7%, 7.0%
Beullens et al.	100	Nominal: 100 in every file	Within it, 2 to 14 routes	5.8%, 5.2%

On 96 of the 97 instances in the first two rows the file gives the fewest vehicles the total demand allows. The 65 extra-vehicle costs compare fairly only with solutions that may also use extra vehicles, and they would need a further check against a fixed fleet. Extra vehicles do not appear to help the method: the instances that need them are the ones where it is furthest from the best published cost, probably because they are also the tightest. The best published solutions come from decades of dedicated metaheuristics and exact methods; the point here is how far the single tour of 1997 already goes when it is simply cut into vehicle loads.

10.6 The windy postman

On a slope or against the wind, a street costs more one way than the other. In the windy postman problem every street can be driven both ways, each at its own cost. The undirected and mixed problems are special cases, with equal costs or with one direction forbidden. It is NP-hard (Guan, 1984), but polynomial when every junction already has even degree: on such Eulerian networks the linear relaxation has an integral optimum (Win, 1989).

The model is that of section 9.2, with a separate cost for each direction. The undirected CPP with every street at its cheaper direction is a further lower bound, since any tour pays at least that for each pass (`chinese_postman/windy.py`).

- **Heuristic** (`windy.solve_win`). Win (1989): solve the undirected CPP on the average of the two costs, then solve the windy postman exactly on the Eulerian network that results. It is never more than twice the optimum.
- **Local search** (`windy.solve_heuristic`). The orientation, rebalancing and reversals of section 9.3, with the cost difference between directions added to the estimate of each reversal.
- **Combined method** (`windy.solve_windy`). Heuristic and bounds first, then branch-and-cut told to beat the heuristic. If it proves that nothing does, the heuristic tour is optimal.

Published instances. The tests used the 120 windy instances of Corberán, Oswald, Plana, Reinelt and Sanchis (2012), read unchanged: 500 to 3,000 junctions, half random graphs (WA) and half imitating street networks (WB), with each direction perturbed by about 10%, 20% or 50% of the mean cost. Their instance page, read on 3 November 2026, lists 99 optima and bounds for the other 21 (`results/wpp_published_values.csv`). Win's method ran on all 120 in 0.1–3.2 seconds each, on a machine also running the next test (`results/windy_win.csv`).

Asymmetry	Instances with a published optimum	Win above the optimum (mean, worst)
About 10%	36	0.01%, 0.03%
About 20%	33	0.05%, 0.10%
About 50%	30	0.27%, 0.44%
All	99	0.10%, 0.44%

It reaches the optimum exactly on only 4 of the 99, yet it is far inside its guarantee of twice the optimum. The gap grows with asymmetry, and it changes little with street density (a mean of 0.09–0.11%), with the family or with size. The local search did not improve on it in any of the runs below.

On 10 of the 21 open instances Win's tour improves on the list: on 6 it is below the listed upper bound (WB2052, WB3031, WB3032, WB3035, WB3051, WB3052), and on the other 4 the list gives none (WA3065, WB3055, WB3061, WB3062). All ten tours were recomputed

independently from the instance files. The list may be behind the literature, so these are comparisons with that list only.

The combined method, with a limit of 2 minutes, was run on all 48 instances of 500 and 1,000 junctions and on every fourth file of the larger sizes. The run reached its 2-hour limit before five of the six instances of 3,000 junctions, so 61 of the planned 66 finished (`results/windy_published.csv`). It proved the optimum on 5 of the 500-junction instances and on no others, and each equals the published value. Branch-and-cut improved on Win's tour on 10 instances. The best tours average 0.09% above the best published value, with a worst case of 0.37%. HiGHS is therefore a weak prover here, where the authors' dedicated branch-and-cut uses cuts of its own for this problem. On these instances the heuristic does the work and the proof remains the hard part.

The browser solver offers a hilly town, with a synthetic surface and costs that follow the slope, solved by Win's method and proved by branch-and-cut on request.

10.7 Every network and every service

Sections 10.3 and 10.4 work on two-way networks. Any of the four kinds of network can also need service on only some streets, or be shared among several postmen, which gives six further problems. All six are NP-hard, since the two-way versions are. The model is that of section 10.6 with two changes: a street without service may be left undriven, and every piece of the streets to serve must be reached from outside (`chinese_postman/windyrural.py`). Those connectivity constraints are added between integer solves, as in section 10.3.

- **Bound.** The linear relaxation with connectivity cuts.
- **Heuristic.** Three starts orient the streets to serve: at their cheaper direction, from the relaxation, and, after the idea of Benavent, Corberán, Piñana, Plana and Sanchis (2005), from an undirected rural postman on the summed costs, followed by a windy postman on the network it gives. Each start is rebalanced by minimum-cost flow, and the pieces are joined by the shortest path that gives the lowest rebalanced total. No approximation ratio is proved.
- **Combined method** (`windyrural.solve`). Heuristic and bound first, then branch-and-cut told to beat the heuristic. Each integer solution that falls into pieces is joined into a valid route, and a better route lowers the cutoff, so a run stopped by its time limit still improves on the heuristic.

- **Several postmen** (`chinese_postman/kpostmen.py`). As in section 10.4: one route is cut into k by dynamic programming, here from about eight opening points, using directed distances. Links then move out of the longest route, with each route reordered after a move, and random moves of short runs of links restart the search. The lower bound is the larger of the farthest single job and a k-th of the bound for one route.

Rural postman, published instances. Three families were read unchanged: the windy rural instances of the page of Corberán, Plana and Sanchis (see also their 2007 paper), with 522 published values (`results/wrpp_published_values.csv`), the mixed rural instances of the same page, and the 61 directed instances of Campos and Savall (1995) as distributed with OAR Lib. Every eighth windy file and all mixed and directed files ran with a limit of 20 seconds for branch-and-cut (`results/rural_general.csv`). Twelve mixed files ran again with 5 minutes (`results/rural_mixed_long.csv`). The limit does not cover the heuristic, so the largest cases took up to 9 minutes.

Network	Instances	Streets	Limit	Optimum proved	Best route above the bound (mean, worst)
One-way	61	211–801	20 s	45	0.74%, 8.9%
One-way, the 16 left open	16	211–792	5 min	14	0.06%, 0.53%
Mixed	36	884–2,678	20 s	2	9.0%, 25%
Mixed	6	884–1,326	5 min	6	0
Mixed	6	1,867–2,678	5 min	0	2.1%, 4.8%
Windy	66	14–3,037	20 s	44	5.8%, 46%

On the 47 windy files of the sample with a published value, all optima, the best route is a mean 0.62% above the published value, with a worst case of 7.6%, and equals it on 32. The heuristic alone is a mean 1.66% above. The other 19 files of the sample are not in the list. No result falls below a published value. All 36 mixed files have a published optimum (`results/mrpp_published_values.csv`, copied from `MRPP.txt` on the instance page), and every value proved here equals it. With 20 seconds the best route is a mean 2.67% above the optimum, with a worst case of 12.4%, and equals it on 2. With 5 minutes it equals it on the 6 files of up to 1,326 streets, and is a mean 1.51% above on the 6 larger ones, with a worst case of 3.4%. No route or bound crosses a published value. The 61 directed files have no published

values in this project, so their test is the bound, and the related published sets below. With 5 minutes 14 of the 16 left open were proved (`results/rural_directed_long.csv`), which makes 59 of 61. The heuristic alone is weak on these two families: on the 45 directed cases proved in 20 seconds it is a mean 8.4% above the optimum, with a worst case of 20%, and on the mixed files a mean 2.7%, with a worst case of 12%. Branch-and-cut does the work there.

The 25% of the 20-second table is mostly a weak bound: the same routes are at most 12.4% above the optimum. On the six files of up to 1,326 streets, 1 to 4 minutes proved the optimum.

Directed general routing and stacker crane. Two related directed sets from the same page do have published values (`results/dgrp_published_values.csv`, `results/scp_published_values.csv`). In the directed general routing problem some vertices must also be visited, which `RuralProblem.vertices` adds as further pieces. The stacker crane problem is a directed rural postman whose arcs to serve rarely touch. These are `results/directed_published.csv` and `results/dgrp_lap.csv`.

The 36 DG files list no vertices to visit. Read literally they give routes below the published optima, and with every vertex required they give those optima, so that reading was adopted. It is an inference from the values, not a statement in the files.

Set	Files	Arcs	Branch-and-cut limit	Proved, each equal to the published optimum	Best route above the optimum (mean, worst)
DG	36	1,166–3,177	60 s	31	1.45%, 18.2%
Stacker crane, random	14	3,824–59,901	60–120 s	14	0
Stacker crane, grid, 100 arcs to serve	10	2,410–3,827	60 s	8	0.47%, 3.0%
Stacker crane, grid, 300 arcs to serve	10	8,625–13,246	60 s	0	8.05%, 13.9%
Stacker crane, grid, 500 arcs to serve	10	12,218–16,491	60 s	0	2.42%, 5.4%
Directed general routing, large	70	50,000	120 s	60	0 on those 60; no route on the other 10

No route falls below a published optimum, no bound rises above one, and no value proved here differs from a published one. The method is weakest when the pieces are many and small, as on the grid files. The random crane files took at most 18 seconds each. The 60 large files proved

took a median of 6.5 seconds, on networks of 5,000 junctions. Above 10,000 links `solve` now runs branch-and-cut first, because on these networks the relaxation is often tight and the joining heuristic is slow. HiGHS sometimes overran its limit, by up to 23 minutes, so the last 23 large files had a hard limit of 5 minutes. It stopped 3 of them, which `results/dgrp_lap.csv` records with no route and the bound of the relaxation. The other 7 of the 10 unproved files ended their time limit with no connected route.

Sixty of the 104 crane files were left out: all 40 of the 100 × 100 grid and the 20 of the 50 × 50 grid with 1,000 and 2,000 arcs to serve. Each has a header that counts more arcs than the file holds (16,269 against 15,535 on `Crane_grid50_1000_8_1`, for instance). On all 20 of the 50 × 50 files the linear bound is above the published optimum, which a complete network could not give, so the files are probably incomplete. This has not been confirmed with the authors.

Several postmen, published optima. The 144 instances of Benavent, Corberán, Plana and Sanchis (2009), with 2 to 6 postmen, give 705 cases: 629 with a published optimum and 76 open (`results/mmwrpp_published_values.csv`, `results/k_published.csv`). They have 10–184 links. The files do not name a depot, so vertex 1 was assumed. The assumption is not verified, but it fits: no route falls below a published optimum or bound, and 340 equal an optimum exactly. With 2 seconds of local search per case the heuristic took at most 5.2 seconds.

Postmen	Published optima	Split alone above the optimum (mean)	After local search (mean, worst)	Equal to the optimum
2	144	8.85%	2.68%, 11.9%	56
3	130	9.31%	2.64%, 11.7%	60
4	124	7.57%	2.20%, 16.0%	65
5	124	6.95%	1.48%, 10.6%	82
6	107	5.47%	1.55%, 17.2%	77
All	629	7.74%	2.15%, 17.2%	340

On the 76 open cases it improves on no published upper bound, staying a mean 6.5% above. These are heuristic results and the worst cases are far from the optimum, so this method gives a good route quickly, without any promise on a given network.

Several postmen, one-way and mixed. No published values were available for these, so each was measured against the lower bound with 2 to 5 postmen and 5 seconds of local search (`results/k_general.csv`). The run reached its 2-hour limit after 8 directed and 3 mixed

instances, of 211–1,318 links, and before the windy ones, which the table above covers. The longest route is a mean 32% above the bound on directed instances and 35% on mixed ones, with worst cases of 52% and 64%. The windy results suggest that much of this distance belongs to the bound, which assumes that the work divides perfectly among the postmen, but on these networks that is not proved. On the mixed instances the local search improved on the split by less than 1% in the time given.

The website solves these problems in the browser with a simpler heuristic in JavaScript, and for one postman offers the proof of this section through Python running in the page.

10.8 Summary

Problem	Complexity	Exact method here	Heuristic and guarantee	Measured distance from the optimum
Every street, two-way or one-way	Polynomial	Matching or minimum-cost flow, in polynomial time	Not needed	0
Every street, first pass dearer	Polynomial	Matching on the travel costs	Not needed	0
Every street, mixed	NP-hard	Branch-and-cut, all benchmark networks and towns of up to 400 junctions; most up to 1,600	Orientation, balancing and local search	Mean 0.41%, worst 4.8% (121 benchmarks and towns); mean 2.09%, worst 5.8% (29 published instances)
Every street, a cost for each way (windy postman)	NP-hard; polynomial if Eulerian	Branch-and-cut, 5 of 61 published instances in two minutes	Win, at most 2; local search	Mean 0.10%, worst 0.44% (99 published instances)
Some streets (rural postman)	NP-hard; polynomial with one piece	Branch-and-cut, all 57 benchmark networks and 10 of the 12 towns of up to 400 junctions	Frederickson, at most 3/2; local search	Mean 0.66%, worst 4.8% (16 proved cases)
k postmen, min-max	NP-hard	Branch-and-cut, small networks	Best split of an optimal route, at most 2 − 1/k; local search	Mean 0.02%, worst 1.3% (62 small cases)

Problem	Complexity	Exact method here	Heuristic and guarantee	Measured distance from the optimum
Some streets, on one-way, mixed or windy networks	NP-hard	Branch-and-cut: 45 of 61 one-way in 20 s; 6 of 12 mixed in 5 min; 44 of 66 windy in 20 s	Orientation, rebalancing and joining; no approximation ratio proved	Best route mean 0.62%, worst 7.6% (47 published windy instances, 20 s); mean 0.76%, worst 3.4% (12 published mixed instances, 5 min); 113 of 150 related directed instances proved, each equal to the published optimum
k postmen, on one-way, mixed or windy networks	NP-hard	Not attempted	Split and local search; no approximation ratio proved	Mean 2.15%, worst 17.2% (629 published optima, depot assumed)
Capacitated fleet (CARP)	NP-hard	Not attempted	Ulusoy split and relocations	Median 5.8% above the best known, number of vehicles free

10.9 Still open

Some variants remain for future work: turn penalties, where U-turns and left turns cost extra and which turn the problem into node routing on a larger graph; hierarchical service, where some streets must be served before others; and several depots.

Real street data. OpenStreetMap now provides free, worldwide street networks, and section 9.4 measures five of them. The website of section 6.6 already reads them: it downloads a district's streets through the Overpass API, keeps junctions and street lengths, marks one-way streets and main roads, so a district such as the ones studied in 1997 can be modelled in seconds instead of digitised by hand. For whole towns, the OSMnx library (Boeing, 2017) downloads the same data as a NetworkX multigraph that `chinese_postman` takes after a short conversion.

11. Conclusions

The Chinese Postman Problem has been solvable in polynomial time since 1973, and today's implementations make that practical on any network a single service crew would plausibly cover. The 2026 version keeps the method of 1997 (pair the odd vertices by shortest paths, duplicate those paths, walk the Eulerian circuit) and improves every stage of it.

1. **Exact matching.** Edmonds' blossom algorithm, through NetworkX, gives the proven optimum on every instance tested: 4 classic graphs, 92 benchmark networks, 53 of them

from real roads in the UK and Belgium, and 15 synthetic towns.

2. **Practical size.** With the LEMON backend, a 40,000-junction network with 12,066 odd vertices solves in a median 110 seconds, and one with 19,942 in 250 seconds; in pure Python the first time covers about 750 odd vertices. Every benchmark network solves in under 2.5 seconds. Every optimum of the fast backend carries a checked dual certificate, and a second blossom code gives the same ten optima.
3. **Directed streets.** The directed CPP, proposed as future work in 1997, is solved by minimum-cost flow in under half a second on a 3,600-junction one-way grid.
4. **A bound for fleets that depends on capacity.** On 63 benchmark instances where every street needs service, the single-postman tour costs a median 95.1% of the best known multi-vehicle solution, from 100% down to 47.5%. The CPP is a quick estimate and a close lower bound when vehicles have capacity to spare, and a weak one when they do not.
5. **Mixed streets.** The mixed CPP, NP-hard, is solved exactly by branch-and-cut on every benchmark network (at most 5.8 s), on all 20 towns of up to 400 junctions and on 8 of the 10 of 900 and 1,600 junctions. Beyond that, five minutes of branch-and-cut end within about 1% of a proven bound, and a polynomial heuristic with dual-guided local search gives a tour 1% to 2.4% above that result in about a minute. On a published test set the optima proved here all equal the published ones, but the picture for the heuristic is less favourable: it is up to 5.8% above the optimum where few streets are one-way, and slower. On five real street networks every optimum is proved in at most 34 seconds, and the heuristic is at most 3.1% above it. Separation and bounding, the core of the 1997 work, is exactly what makes this possible.
6. **The paths left open in 1997.** A first pass dearer than the next keeps the problem polynomial when every street needs service. With a different cost each way (the windy postman, NP-hard), Win's method comes within a mean 0.10% of the 99 published optima in seconds. When only some streets do (the rural postman problem, NP-hard), branch-and-cut proves the optimum on all 57 benchmark networks that mark them, and the heuristic averages 0.66% above it. Several postmen sharing a district (NP-hard) are planned by the best split of one tour and a local search, optimal in 61 of 62 small cases where the optimum is known; with vehicle capacities, the same idea comes within a median 5.8% of the best published costs, with the number of vehicles left free. The six remaining combinations of network and service, rural and several postmen on one-way, mixed and windy networks, are also solved (section 10.7): on 47 published windy rural instances the best route is a mean 0.62% above the published optimum, and on 629 published min-max

optima the several-postmen heuristic is a mean 2.15% above, with worst cases of 7.6% and 17.2%.

7. **Open to everyone.** The code, the results, this report and the 1997 original are published at github.com/miquelmp/chinese-postman-problem, and the website solves street networks of up to 2,500 junctions in the browser, from a file or from OpenStreetMap, in seconds.

The two observations of 1997 about practice still hold. Routine services such as refuse collection or mail delivery are bounded by a working shift, so the networks involved stay moderate in size; and for one-off emergencies such as snow clearing, a fast near-optimal answer can be worth more than a slow proven one. For the CPP itself, thanks to the matching theory of the 1960s and 1970s and to today's implementations, a fast answer and a proven one are the same answer.

The natural continuation is a stronger lower bound for several postmen, real slopes for the windy postman and turn penalties for real vehicles, all on the same code base, the same benchmark files and more real street networks from OpenStreetMap.

References

Problem and algorithms

- Euler, L. (1741). Solutio problematis ad geometriam situs pertinentis. *Commentarii academiae scientiarum Petropolitanae* 8, 128–140.
- Dijkstra, E. W. (1959). A note on two problems in connexion with graphs. *Numerische Mathematik* 1, 269–271.
- Kwan Mei-Ko (1960). Graphic programming using odd or even points. *Acta Mathematica Sinica* 10, 263–266 (in Chinese). English translation in *Chinese Mathematics* 1 (1962), 273–277.
- Edmonds, J. (1965). Paths, trees, and flowers. *Canadian Journal of Mathematics* 17, 449–467.
- Edmonds, J. (1965). Maximum matching and a polyhedron with 0,1-vertices. *Journal of Research of the National Bureau of Standards* 69B, 125–130.
- Edmonds, J. and Johnson, E. L. (1973). Matching, Euler tours and the Chinese postman. *Mathematical Programming* 5, 88–124.

- Hierholzer, C. (1873). Über die Möglichkeit, einen Linienzug ohne Wiederholung und ohne Unterbrechung zu umfahren. *Mathematische Annalen* 6, 30–32.
- Papadimitriou, C. H. (1976). On the complexity of edge traversing. *Journal of the ACM* 23(3), 544–554.
- Frederickson, G. N. (1979). Approximation algorithms for some postman problems. *Journal of the ACM* 26(3), 538–554.
- Cook, S. A. (1971). The complexity of theorem-proving procedures. *Proceedings of the 3rd Annual ACM Symposium on Theory of Computing*, 151–158.
- Levin, L. A. (1973). Universal sequential search problems. *Problemy Peredachi Informatsii* 9(3), 115–116 (in Russian).
- Clay Mathematics Institute (2000). The Millennium Prize Problems: P vs NP. claymath.org/millennium/p-vs-np

Matching algorithms

- Gabow, H. N. (1973). *Implementations of algorithms for maximum matching on nonbipartite graphs*. PhD thesis, Stanford University.
- Lawler, E. L. (1976). *Combinatorial Optimization: Networks and Matroids*. Holt, Rinehart and Winston, New York.
- Micali, S. and Vazirani, V. V. (1980). An O(√|V|·|E|) algorithm for finding maximum matching in general graphs. *Proceedings of the 21st Annual Symposium on Foundations of Computer Science*, 17–27.
- Galil, Z., Micali, S. and Gabow, H. (1986). An O(EV log V) algorithm for finding a maximal weighted matching in general graphs. *SIAM Journal on Computing* 15(1), 120–130.
- Gabow, H. N. (1990). Data structures for weighted matching and nearest common ancestors with linking. *Proceedings of the 1st Annual ACM-SIAM Symposium on Discrete Algorithms*, 434–443.
- Gabow, H. N. and Tarjan, R. E. (1991). Faster scaling algorithms for general graph matching problems. *Journal of the ACM* 38(4), 815–853.
- Cook, W. and Rohe, A. (1999). Computing minimum-weight perfect matchings. *INFORMS Journal on Computing* 11(2), 138–148.
- Mehlhorn, K. and Schäfer, G. (2002). Implementation of O(nm log n) weighted matchings in general graphs: the power of data structures. *ACM Journal of Experimental Algorithmics* 7, article 4.

- Kolmogorov, V. (2009). Blossom V: a new implementation of a minimum cost perfect matching algorithm. *Mathematical Programming Computation* 1(1), 43–67.
- Duan, R. and Pettie, S. (2014). Linear-time approximation for maximum weight matching. *Journal of the ACM* 61(1), article 1.
- Duan, R., Pettie, S. and Su, H.-H. (2018). Scaling algorithms for weighted matching in general graphs. *ACM Transactions on Algorithms* 14(1), article 8.
- Chen, L., Kyng, R., Liu, Y. P., Peng, R., Probst Gutenberg, M. and Sachdeva, S. (2022). Maximum flow and minimum-cost flow in almost-linear time. *Proceedings of the 63rd IEEE Symposium on Foundations of Computer Science*, 612–623.
- Wu, Y. and Zhong, L. (2023). Fusion Blossom: fast MWPM decoders for QEC. *Proceedings of the IEEE International Conference on Quantum Computing and Engineering*. arXiv:2305.08307.
- Higgott, O. and Gidney, C. (2025). Sparse Blossom: correcting a million errors per core second with minimum-weight matching. *Quantum* 9, 1600.
- Arkhipov, P. and Kolmogorov, V. (2026). Blossom VI: a practical minimum weight perfect matching algorithm. arXiv:2604.20351.

Arc routing

- Corberán, Á. and Laporte, G., eds. (2014). *Arc Routing: Problems, Methods, and Applications*. SIAM, Philadelphia.
- Eiselt, H. A., Gendreau, M. and Laporte, G. (1995). Arc routing problems, part I: the Chinese postman problem. *Operations Research* 43, 231–242.
- Lenstra, J. K. and Rinnooy Kan, A. H. G. (1976). On general routing problems. *Networks* 6(3), 273–280.
- Frederickson, G. N., Hecht, M. S. and Kim, C. E. (1978). Approximation algorithms for some routing problems. *SIAM Journal on Computing* 7(2), 178–193.
- Benavent, E., Campos, V., Corberán, Á. and Mota, E. (1985). Análisis de heurísticos para el problema del cartero rural. *Trabajos de Estadística y de Investigación Operativa* 36, 27–38.
- Golden, B. L. and Wong, R. T. (1981). Capacitated arc routing problems. *Networks* 11(3), 305–315.
- Bode, C. and Irnich, S. (2012). Cut-first branch-and-price-second for the capacitated arc-routing problem. *Operations Research* 60(5), 1167–1182.
- Chen, Y., Hao, J.-K. and Glover, F. (2016). A hybrid metaheuristic approach for the capacitated arc routing problem. *European Journal of Operational Research* 253(1), 25–

39.
- Guan, M. G. (1984). On the windy postman problem. *Discrete Applied Mathematics* 9(1), 41–46.
- Win, Z. (1989). On the windy postman problem on Eulerian graphs. *Mathematical Programming* 44(1), 97–112.
- Campos, V. and Savall, J. V. (1995). A computational study of several heuristics for the DRPP. *Computational Optimization and Applications* 4(1), 67–77.
- Benavent, E., Corberán, Á., Piñana, E., Plana, I. and Sanchis, J. M. (2005). New heuristic algorithms for the windy rural postman problem. *Computers & Operations Research* 32(12), 3111–3128.
- Corberán, Á., Plana, I. and Sanchis, J. M. (2007). A branch & cut algorithm for the windy general routing problem and special cases. *Networks* 49(4), 245–257.
- Benavent, E., Corberán, Á., Plana, I. and Sanchis, J. M. (2009). Min-max K-vehicles windy rural postman problem. *Networks* 54(4), 216–226.
- Ulusoy, G. (1985). The fleet size and mix problem for capacitated arc routing. *European Journal of Operational Research* 22(3), 329–337.
- Ghiani, G. and Laporte, G. (2000). A branch-and-cut algorithm for the undirected rural postman problem. *Mathematical Programming* 87(3), 467–481.

Benchmark instances

- Corberán, Á., Oswald, M., Plana, I., Reinelt, G. and Sanchis, J. M. (2012). New results on the Windy Postman Problem. *Mathematical Programming* 132(1), 309–332. Source of the MA and MB instances of section 9.4.
- Corberán, Á., Plana, I. and Sanchis, J. M. Arc routing problems: data instances, uv.es/corberan/instancias.htm. Source of the published values of section 9.4.
- Golden, B. L., DeArmon, J. S. and Baker, E. K. (1983). Computational experiments with algorithms for a class of routing problems. *Computers & Operations Research* 10, 47–59. (gdb)
- Benavent, E., Campos, V., Corberán, Á. and Mota, E. (1992). The capacitated arc routing problem: lower bounds. *Networks* 22, 669–690. (val)
- Li, L. Y. O. and Eglese, R. W. (1996). An interactive algorithm for vehicle routeing for winter-gritting. *Journal of the Operational Research Society* 47, 217–228. (egl)
- Beullens, P., Muyldermans, L., Cattrysse, D. and Van Oudheusden, D. (2003). A guided local search heuristic for the capacitated arc routing problem. *European Journal of Operational Research* 147, 629–643. (C–F)

- Brandão, J. and Eglese, R. (2008). A deterministic tabu search algorithm for the capacitated arc routing problem. *Computers & Operations Research* 35, 1112–1126. (egl-g)
- Martinelli, R. CARPData.jl: classical CARP instances in a common format. github.com/rafaelmartinelli/CARPData.jl (MIT licence; files used as distributed).
- Kiuchi, M., Shinano, Y., Hirabayashi, R. and Saruwatari, Y. (1995). An exact algorithm for the capacitated arc routing problem using parallel branch and bound method. *Abstracts of the 1995 Spring National Conference of the Operations Research Society of Japan*, 28–29. (kshs)

Software

- Lum, O., Golden, B. and Wasil, E. (2019). OAR Lib: an open source arc routing library. *Mathematical Programming Computation* 11(3), 587–629. Distributes the mixed instances of section 9.4.
- seen-one. Route Crafter, github.com/seen-one/Route-Crafter.
- Brooks, A. W. postman_problems: solutions to the Chinese postman problem and the rural postman problem, github.com/brooksandrew/postman_problems.
- Hagberg, A., Schult, D. and Swart, P. (2008). Exploring network structure, dynamics, and function using NetworkX. *Proceedings of SciPy 2008*, 11–15.
- Virtanen, P. et al. (2020). SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods* 17, 261–272.
- Huangfu, Q. and Hall, J. A. J. (2018). Parallelizing the dual revised simplex method. *Mathematical Programming Computation* 10, 119–142. (HiGHS)
- Dezső, B., Jüttner, A. and Kovács, P. (2011). LEMON – an open source C++ graph template library. *Electronic Notes in Theoretical Computer Science* 264(5), 23–45.
- Van Rantwijk, J. (2008). Maximum weighted matching in general graphs (mwmatching.py), jorisvr.nl/article/maximum-matching. Basis of the JavaScript matching code.
- Boeing, G. (2017). OSMnx: new methods for acquiring, constructing, analyzing, and visualizing complex street networks. *Computers, Environment and Urban Systems* 65, 126–139.
- Pyodide: Python compiled to WebAssembly, pyodide.org. Used for the in-browser optimality proof.
- OpenStreetMap contributors. Street data © OpenStreetMap contributors, Open Database Licence, read through the Overpass API.

Original project

- Monserrat Pons, M. (1997). *Resolución del problema del cartero chino (CPP) mediante la aplicación de un proceso de branch and bound*. Final-year project, ETSEIB, Barcelona. Supervisor: R. Companys Pascual.